

Pumps 2-3 maintenance works

PUMP SERVICE & MAINTENANCE CHECKLIST

Customer:	SIT
End-user:	SIT
Location of site:	L1 CHILLER Plant room
Contact person No:	

Designation of pump system no.:	CHWP 02
Pump model serial No.:	GS150-315 / T210595-04
Motor model power (kW) serial No.:	13007013E

Description		Record			Remark
1.	Motor voltage (Volts)	427.1 428.6 429.0			
2.	Motor running current (Amps)	L1	L2	L3	
		44A	41.8A	42.6A	
3.	Operating pressure (Bar)	Suction		Discharge	
		5.8 bar		6.28 bar	
4.	Flow rate (m ³ /hr)	587.6 m ³ /hr			
5.	Check for Leakage (Mech seal / Gland packing)	Pass		Fail	
6.	Pump & motor mounting check	Pass		Fail	
7.	General pump vibration check	Pass		Fail	
8.	Bearing temperature	37°C			
9.	General cleaning	Completed			

Note: To be recorded only when the date is available on site

Remark:

Mohan  15/01/2025
Technician's Name & Signature

Won  16/11/2025
Verified by Name & Signature

PUMP SERVICE & MAINTENANCE CHECKLIST

Customer:	SIT
End-user:	SIT
Location of site:	L1 CHILLER Plant room
Contact person No:	

Designation of pump system no.:	CHWP-03
Pump model serial No.:	GS150-315 / T210595-03
Motor model power (kW) serial No.:	13007011 E

Description		Record			Remark
1.	Motor voltage (Volts)	428.8 427.3 427.1			
2.	Motor running current (Amps)	L1	L2	L3	
		45.1A	41.6A	47.2A	
3.	Operating pressure (Bar)	Suction		Discharge	
		5.8 bar		6.28 bar	
4.	Flow rate (m ³ /hr)	587.6 m ³ /hr			
5.	Check for Leakage (Mech seal / Gland packing)	Pass		Fail	
6.	Pump & motor mounting check	Pass		Fail	
7.	General pump vibration check	Pass		Fail	
8.	Bearing temperature	36°C			
9.	General cleaning	Completed			

Note: To be recorded only when the date is available on site

Remark:

Mohan  15/01/2025
Technician's Name & Signature

Man  16/1/2025
Verified by Name & Signature

PUMP SERVICE & MAINTENANCE CHECKLIST

Customer:	SIT
End-user:	SIT
Location of site:	LA CHILLER Plant room
Contact person No:	

Designation of pump system no.:	CDWP-02
Pump model serial No.:	350X300CBF
Motor model power (kW) serial No.:	13014001E

Description		Record			Remark
1.	Motor voltage (Volts)	425.1 425.6 426.1			
2.	Motor running current (Amps)	L1	L2	L3	
		45.2A	46.1A	43.9A	
3.	Operating pressure (Bar)	Suction	Discharge		
		5.9 bar	5.95 bar		
4.	Flow rate (m ³ /hr)	518 m ³ /bar			
5.	Check for Leakage (Mech seal / Gland packing)	Pass	Fail		
6.	Pump & motor mounting check	Pass	Fail		
7.	General pump vibration check	Pass	Fail		
8.	Bearing temperature	36°C			
9.	General cleaning	Completed			

Note: To be recorded only when the date is available on site

Remark:

Mohan  15/01/2025
Technician's Name & Signature

Ivan  16/1/2025
Verified by Name & Signature

PUMP SERVICE & MAINTENANCE CHECKLIST

Customer:	SIT
End-user:	SIT
Location of site:	LI CHILLER Plant room
Contact person No:	

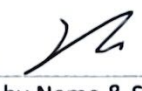
Designation of pump system no.:	CDWP-03
Pump model serial No.:	350 x 300 CBF
Motor model power (kW) serial No.:	13014 1005E

Description		Record			Remark
1.	Motor voltage (Volts)	428.1 427.4 426.9			
2.	Motor running current (Amps)	L1	L2	L3	
		48.8 A	46.1 A	45.7 A	
3.	Operating pressure (Bar)	Suction	Discharge		
		5.9 bar	5.95 bar		
4.	Flow rate (m ³ /hr)	518 m ³ /hr			
5.	Check for Leakage (Mech seal / Gland packing)	Pass	Fail		
6.	Pump & motor mounting check	Pass	Fail		
7.	General pump vibration check	Pass	Fail		
8.	Bearing temperature	36.1 °C			
9.	General cleaning	Completed			

Note: To be recorded only when the date is available on site

Remark:

Mohan  15/01/2025
Technician's Name & Signature

Ivan  16/1/2025
Verified by Name & Signature

Cooling towers 2-3 maintenance works

Cooling Tower #2

Cooling Tower System Monthly Service report
OperationPeriod (Please circle): Startup / Shutdown / Monthly / Quarterly

Service type	Results: Tick (✓) if ok	Observation & Comments
Strainer & water sump is clear for flowing	✓	
Cold water basin & hot water basin no blocking	✓	
Air inlet louvers & fan guards in good condition	✓	
Water level in cold water basin as per design	✓	
Check fan for cracks or scale buildup	✓	
Check the condition of belt (i.e. wear and tear)	✓	
Check belt tension (i.e. Loosen or sounds)	✓	
Check the Make-up valve	✓	
Lubricate fan shaft bearings (any abnormal sounds)	✓	
Check motor for unusual noise/smell	✓	
Inspect casing and basin for any cracks and leakage	✓	
SSF running record from BMS	✓	
SSF discharge pressure in pump side	✓	
SSF no alarm records	✓	

Remark:

Mohan  15/01/2025
Technician's Name & Signature

Ivan  16/1/2025
Verified by Name & Signature

Cooling tower #3

Period (Please circle): Startup / Shutdown / Monthly / Quarterly

Service type	Results: Tick (✓) if ok	Observation & Comments
Strainer & water sump is clear for flowing	✓	
Cold water basin & hot water basin no blocking	✓	
Air inlet louvers & fan guards in good condition	✓	
Water level in cold water basin as per design	✓	
Check fan for cracks or scale buildup	✓	
Check the condition of belt (i.e. wear and tear)	✓	
Check belt tension (i.e. Loosen or sounds)	✓	
Check the Make-up valve	✓	
Lubricate fan shaft bearings (any abnormal sounds)	✓	
Check motor for unusual noise/smell	✓	
Inspect casing and basin for any cracks and leakage	✓	
SSF running record from BMS	✓	
SSF discharge pressure in pump side	✓	
SSF no alarm records	✓	

Remark:

 Mohan  15/01/2025
 Technician's Name & Signature

 War 16/1/2025
 Verified by Name & Signature

Chillers 2-3 maintenance works

Service Company: <u>ENGIE</u>	Site Name: <u>SIT</u>
Date: <u>15/01/2025</u>	Chiller Serial No.: <u>211420 CH-SIT-L1-03</u>

Items	Status
Refrigeration	
1. Leaking sensor is monitoring (running).	(Pass) Fail
2. Liquid level sensor and EXVs in normal	(Pass) Fail
Compressors:	
1. Comm signal is working as normal.	(Pass) Fail
2. In/out TEMP reading	Discharge: <u>42.1</u> Suction: <u>11.2</u>
3. Date and time via the SMT	SMT: <u>20:18</u> GMT+8: <u>20:25</u>
4. Abnormal checking	(Pass) Fail

Evaporator	Condenser
CHW Inlet Temp. (°C) <u>9.6</u>	CW Inlet Temp. (°C) <u>28.9</u>
CHW Outlet Temp. (°C) <u>7.6</u>	CW Outlet Temp. (°C) <u>32.8</u>
Evap Split(k) <u>277.2</u>	CW Split(k) <u>277.8</u>
Evap Pressure(kPa) <u>263.3</u>	CW Pressure(kPa) <u>765.6</u>
Saturated Evap Temp. (°C) <u>7.1</u>	Saturated Evap Temp. (°C) <u>34.1</u>
Evap Approach Temp. (°C) <u>0.3</u>	Cond Approach Temp. (°C) <u>0.3</u>
Design Evap Pressure Drop(kPaD) <u>45.1</u>	Discharge Temp. (°C) <u>43.2</u>
Design Evap Flow Rate(l/s) <u>181.8</u>	Discharge Superheat (k) <u>283.5</u>
Actual Evap Pressure Drop(kPaD) <u>32.3</u>	Liquid Line Temp. (°C) <u>29.5</u>
Actual Evap Flow Rate(l/s) <u>161.5</u>	Liquid Sub-Cooling (k) <u>278.4</u>
	Sight Glass (Clear or Flashing) <u>CLEAR</u>
	Design Cond Pressure Drop(kPaD) <u>50</u>
	Design Cond Flow Rate(l/s) <u>228.7</u>
	Actual Cond Pressure Drop(kPaD) <u>N/A</u>
	Actual Cond Flow Rate(l/s) <u>165.7</u>

Remark:

Moham 15/01/2025
Technician's Name & Signature

16/01/2025
Verified by Name & Signature

Service Company: <u>ENGIE</u>	Site Name: <u>SIT</u>
Date: <u>15/01/2025</u>	Chiller Serial No.: <u>211419 CH-SIT-L1-02</u>

Items	Status
Refrigeration	
1. Leaking sensor is monitoring (running).	<u>Pass</u> Fail
2. Liquid level sensor and EXVs in normal	<u>Pass</u> Fail
Compressors:	
1. Comm signal is working as normal.	<u>Pass</u> Fail
2. In/out TEMP reading	Discharge: <u>37.5</u> Suction: <u>11.2</u>
3. Date and time via the SMT	SMT: <u>19:55</u> GMT+8: <u>20:09</u>
4. Abnormal checking	<u>Pass</u> Fail

Evaporator	Condenser
CHW Inlet Temp. (°C) <u>9.5</u>	CW Inlet Temp. (°C) <u>28.1</u>
CHW Outlet Temp. (°C) <u>6.7</u>	CW Outlet Temp. (°C) <u>31.8</u>
Evap Split(k) <u>277.1</u>	CW Split(k) <u>277.2</u>
Evap Pressure(kPa) <u>263.7</u>	CW Pressure(kPa) <u>720.6</u>
Saturated Evap Temp. (°C) <u>6.2</u>	Saturated Evap Temp. (°C) <u>32.2</u>
Evap Approach Temp. (°C) <u>0.4</u>	Cond Approach Temp. (°C) <u>0.3</u>
Design Evap Pressure Drop(kPaD) <u>34.5</u>	Discharge Temp. (°C) <u>37.5</u>
Design Evap Flow Rate(l/s) <u>181.8</u>	Discharge Superheat (k) <u>28.6</u>
Actual Evap Pressure Drop(kPaD) <u>48.7</u>	Liquid Line Temp. (°C) <u>23</u>
Actual Evap Flow Rate(l/s) <u>143.5</u>	Liquid Sub-Cooling (k) <u>277.6</u>
	Sight Glass (Clear or Flashing) <u>CLEAR</u>
	Design Cond Pressure Drop(kPaD) <u>50</u>
	Design Cond Flow Rate(l/s) <u>228.7</u>
	Actual Cond Pressure Drop(kPaD) <u>N.A</u>
	Actual Cond Flow Rate(l/s) <u>156.7</u>

Remark:

Moham ALM 15/01/2025
Technician's Name & Signature

22
Ivan 16/1/2025
Verified by Name & Signature

Chemical maintenance works

Autotube system maintenance works

Service Company: <u>ENGIE</u>	Site Name: <u>SIT</u>
Date: <u>15/01/2025</u>	Chiller system: <u>CH-SIT-L1-02</u>

Items	Description	Status (Pass / Fail)
1.	Check the system for sign of leak	Pass
2.	Record the clocking meter	Pass
3.	Clean the sight glass	Pass
4.	Check the pump for abnormal vibration	Pass
5.	Check the tightness of foundation bolts of pump	Pass
6.	Check the Electrical system working well (no alarm)	Pass

Remark:

Mohan 15/01/2025
Technician's Name & Signature

Ivan 16/1/2025
Verified by Name & Signature

BMS system maintenance works



Singapore Institute of Technology

DLP Maintenance Checklist

TITLE : DLP Preventive Maintenance for EMS System
FREQUENCY : Monthly
DATE/TIME : 13 Jan 2025 – 15 Jan 2025

1) Inspection of EMS Workstation

Description of Work: Perform physical, visual, and software level checks as stated below.

Location: Plant Control Room

S/N	Description	Remarks
1.1	Check Computer, Monitor and Printer (if applicable) Operation	A
1.2	Perform Back-up Hard Drive for Server and Control Database	A
1.3	Check Set-up and Operation of BMS Software	A
1.4	Check Workstation Free from Dust and Stain	A
1.5	Network Communication	A
1.6	Alarm and Trend Data Available	A
1.7	Backup Trend Data	A
1.8	Run Anti-Virus Scan (when applicable)	A
1.9	Check UPS (if applicable)	A

LEGEND

A: Serviced

B: Normal

C: Faulty

D: Others (with remarks)



Singapore Institute of Technology

DLP Maintenance Checklist

2) Inspection of DDC Panels

Description of Work: Perform physical, visual, and software level checks as stated below.

Location: Plant Room & Rooftop

DDC No.:		DCS-SIT_DDC-W3-L1-01	DCS-SIT_DDC-W3-L1-02	DCS-SIT_DDC-W3-L10-01	SSF_CHW	SSF_CW
S/N	Description	Remarks				
2.1	Clean and Inspect the Control Panel Free of Dust and Stain	A	A	A	A	A
2.2	Check Exhaust Fan/Filter is in Working Condition (if applicable)	A	A	A	A	A
2.3	Check Fan Thermostat in Working Condition (if applicable)	A	A	A	A	A
2.4	Check Control Panel Lock	A	A	A	A	A
2.5	Check Panel Light Interlock (if applicable)	A	A	A	A	A
2.6	Check Panel Hinges Condition	A	A	A	A	A
2.7	Check HMI Panel (if applicable)	A	A	A	A	A
2.8	Check Energy Meter Display Condition Status	A	A	A	A	A
2.9	Check DDC/PLC Tx/Rx Signal Status	A	A	A	A	A
2.10	Check Gateway Tx/Rx Status	A	A	A	A	A
2.11	Check UPS (if applicable)	A	A	A	A	A
2.12	Check Network Switch	A	A	A	A	A

LEGEND

A: Serviced

B: Normal

C: Faulty

D: Others (with remarks)



Singapore Institute of Technology

DLP Maintenance Checklist

DDC No.:		TRWP	FP	SP	MWP	
S/N	Description	Remarks				
2.1	Clean and Inspect the Control Panel Free of Dust and Stain	A	A	A	A	
2.2	Check Exhaust Fan/Filter is in Working Condition (if applicable)	A	A	A	A	
2.3	Check Fan Thermostat in Working Condition (if applicable)	A	A	A	A	
2.4	Check Control Panel Lock	A	A	A	A	
2.5	Check Panel Light Interlock (if applicable)	A	A	A	A	
2.6	Check Panel Hinges Condition	A	A	A	A	
2.7	Check HMI Panel (if applicable)	A	A	A	A	
2.8	Check Energy Meter Display Condition Status	A	A	A	A	
2.9	Check DDC/PLC Tx/Rx Signal Status	A	A	A	A	
2.10	Check Gateway Tx/Rx Status	A	A	A	A	
2.11	Check UPS (if applicable)	A	A	A	A	
2.12	Check Network Switch	A	A	A	A	

LEGEND

A: Serviced

B: Normal

C: Faulty

D: Others (with remarks)



Singapore Institute of Technology

DLP Maintenance Checklist

Additional Remarks:

Attended by:

Ron Chan
15 Jan 2025

Engineer (ENGIE)
(Name/Signature/Date)

Checked and acknowledge by:

KK Kwan
15 Jan 2025

Engineer/Manager (ENGIE)
(Name/Signature/Date)

Acknowledge by:

16/1/2025

(Name/Signature/Date)

LEGEND

A: Serviced

B: Normal

C: Faulty

D: Others (with remarks)